

5. Gaussian Distribution

Theorem 13. (Single-variable Gaussian) Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then its PDF is

$$f(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

Proof. Not required. ■

Theorem 14. (Multivariate Gaussian) Let $\mathbf{X} \sim \mathcal{N}(\mu, \Sigma) \in \mathbb{R}^d$. Then its PDF is

$$f(\mathbf{x} | \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2} (\mathbf{x} - \mu)^\top \Sigma^{-1} (\mathbf{x} - \mu)\right).$$

Proof. Not required. ■

Theorem 15. Let $\mathbf{X}^{(0)}, \dots, \mathbf{X}^{(N-1)} \in \mathbb{R}^{d \times 1}$ be jointly normal. Define $\mathbf{Y} = \sum_{n=0}^{N-1} c_n \mathbf{X}^{(n)} \in \mathbb{R}^{d \times 1}$. Then

1. $\mathbf{Y}$ is a normal random vector.

2.

$$\mathbb{E}[\mathbf{Y}] = \sum_{n=0}^{N-1} c_n \mathbb{E}[\mathbf{X}^{(n)}]$$

3.

$$\text{Var}[\mathbf{Y}] = \sum_{n=0}^{N-1} c_n^2 \text{Var}[\mathbf{X}^{(n)}] + 2 \sum_{0 \leq n < m \leq N-1} c_n c_m \text{Cov}[\mathbf{X}^{(n)}, \mathbf{X}^{(m)}].$$

Proof. The proof of Part 1 is not required. The proofs of other parts are omitted. ■

Theorem 16. (Marginal Gaussians are Gaussians) Let $\mathbf{X} \in \mathbb{R}^{d_1 \times 1}$, $\mathbf{Y} \in \mathbb{R}^{d_2 \times 1}$, and

$$\begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_{\mathbf{X}} \\ \mu_{\mathbf{Y}} \end{bmatrix}, \begin{bmatrix} \Sigma_{\mathbf{XX}} & \Sigma_{\mathbf{XY}} \\ \Sigma_{\mathbf{YX}} & \Sigma_{\mathbf{YY}} \end{bmatrix}\right)$$

where $\Sigma_{ij} = \text{Cov}[\mathbf{i}, \mathbf{j}]$ for $i, j \in \{\mathbf{X}, \mathbf{Y}\}$.

Then

$$\mathbf{X} \sim \mathcal{N}(\mu_{\mathbf{X}}, \Sigma_{\mathbf{XX}}).$$

Proof. Not required. ■

Theorem 17. (Conditional Gaussians are Gaussians) Let

$$\begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_{\mathbf{X}} \\ \mu_{\mathbf{Y}} \end{bmatrix}, \begin{bmatrix} \Sigma_{\mathbf{XX}} & \Sigma_{\mathbf{XY}} \\ \Sigma_{\mathbf{YX}} & \Sigma_{\mathbf{YY}} \end{bmatrix}\right)$$

where $\Sigma_{ij} = \text{Cov}[\mathbf{i}, \mathbf{j}]$ for $i, j \in \{X, Y\}$.

Then conditional on $\mathbf{Y} = \mathbf{y}$, the conditional distribution of $\mathbf{X}$ is Gaussian:

$$\mathbf{X} | (\mathbf{Y} = \mathbf{y}) \sim \mathcal{N}(\mu_{\mathbf{X}|\mathbf{y}}, \Sigma_{\mathbf{X}|\mathbf{y}}),$$